Audio Beam Steering With Phased Array Method Using Arduino Due Microcontroller

Amaro Da Silva Gaviola, Muhammad Rivai, Hendra Kusuma
Department of Electrical Engineering
Institut Teknologi Sepuluh Nopember
Surabaya,Indonesia
gaviola78marro@gmail.com, muhammad_rivai@ee.its.ac.id, hendraks@ee.its.ac.id

***Abstract*— The audio system requires a large amount of power to transmit sound energy to all areas both inside and outside the room. Sound waves can propagate in the air and undergo great absorption in their propagation. Often people or audiences only exist in areas with a certain distance and direction, so much of the sound energy is wasted. The focusing technique can be applied to transmit sound energy to the audience area with a certain direction and distance. Thus, this technique can save energy significantly. Audio steering system with mechanical process, requires high electric driving power and takes a long time. In this research, a sound energy guiding system from four loudspeaker series has been done with phase difference or phased array method. So, there is no mechanical loudspeaker modifiers are needed. The computer is used as a Graphical User Interface to determine the direction and distance of the audience who will receive the sound, by generating the delay configuration pattern of each path using Arduino Due Microcontroller. So this process can work automatically in directing the sound energy. The test results show that the sound intensity generated by the system is greater at the selected or target angle than the others. Beam Steering audio is expected to be used for large meeting rooms or outdoors, where the sound energy is only sent towards the desired power depending on the area and distance of the audience.**

***Keywords*—audio beam steering, graphical user interface, loudspeaker series, phased array.**

I. Introduction

In general, the laying of the audio loudspeaker in the crowd is placed randomly or a particular location. But, in reality, often people or audiences only exist in areas with a certain distance and direction, so a lot of sound energy wasted. In addition, the laying of the speaker's direction greatly affects the voice conversation, which sometimes the audience cannot hear clearly.

Sound waves can spread in the air media and undergo great absorption in their propagation [1]. Audio systems require a large amount of energy to transmit sound energy to all areas in both indoors and outdoors. The focusing technique can be applied to transmit sound energy to an area of the audience with a certain direction and distance so as to conserve the electrical energy significantly. Directing sound energy mechanically can be done by changing the physical direction of the loudspeaker used. However, this technique requires high electric driving power and takes a long time [2].

Phase Array is a technique of focusing sound waves or Audio Beam Steering (ABS). This method has been carried out research on ABS implemented in classrooms aimed at delivering material that can be hear clearly for all directions of the room [3]. In this research we will develop the focus of the sound waves consisting of a series of audio loudspeakers with a computer used as a Graphical User Interface (GUI) and Arduino Due Microcontroller used as generator of phased array that allows the user to change the number of frequencies and the direction of the sounds as desired.

II. Material and Design

A. Design of loudspeaker arrays

The design of the loudspeaker is arranged using four speakers arranged horizontally. To avoid grating lobes [4], the spacing between speakers can be determined using the Eq. (1),

$$d \leq \frac{\lambda}{1+\sin\theta_{\max}} \tag{1}$$

where λ is the wavelength (m), ɵ is the direction angle (o) and d is the distance between speakers (m).

Since the frequency of the sinusoidal signal is 1 kHz, the wavelength can be calculated using Eq. (2),

$$\lambda = \frac{1}{f} * c \tag{2}$$

where λ is the wavelength (m), f is the frequency (Hz), and c is the speed of sound (343.2 m/s). So, the value of λ can be calculated, as in Eq. (3),

$$\lambda = \frac{1}{1khz} * \frac{343.2m}{s} = 0.3m \tag{3}$$

Therefore, based on the Eq. (1), the distance between speakers is determined as in Eq. (4),

$$d \leq \frac{0.3m}{1+\sin 90} \leq \frac{0.3m}{1+1} \leq 15cm \tag{4}$$

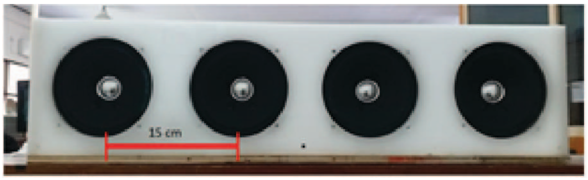


Fig. 1. Loudspeaker array design

The speaker settings are shown in Fig. 1. The time delay of each loudspeaker can be obtained by assuming that the sound is a particle moving from one point to another within a certain time.

B. Delay Calculation Using Basic Geometry

The delay time of each loudspeaker can be determined using the basics of geometry. Four speakers placed along the *x*-axis, as shown in Fig. 2. It reflects the position of four speaker arrays. In this configuration, the transmitting ABS angle is measured on the *y*-axis.

As shown in Fig. 2, *d* is the distance of each speaker and θ is the angle formed by the object to the horizontal lines of the array loudspeaker. If the signal comes out of two speakers coherently or exits simultaneously with the same phase angle then the signal will be amplified in a straight direction in front of the speaker [5]. If the direction desired by the user is not straight, then there will be different wavelengths of sound coming out of the speakers. Thus a delay is given to one of the speakers, so that the signal on the speaker can start coherently with the other speakers at the end of the time delay [6], as shown in Fig. 3.

The calculation of the time delay required by each loudspeaker [7], can be defined in Eq. (5),

$$Delay = \frac{Distance}{Speed\ of\ sound} = \frac{d \sin \theta}{c} \qquad (5)$$

where d is a distance of loudspeakers, θ is an angle of direction and c is the speed of sound in the air.

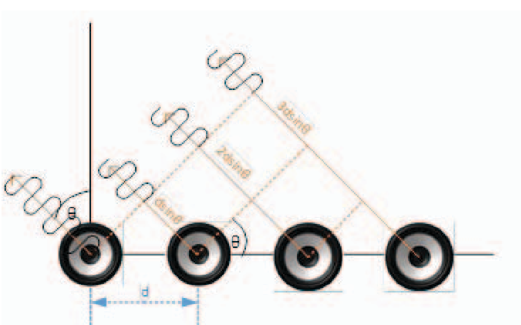


Fig. 2. Geometry of sound source series

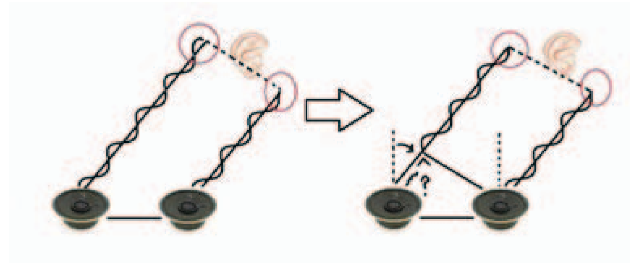


Fig. 3. Delay calculation

C. Graphical User Interface Design

Processing Foundation software is used in GUI design. In the GUI, the user can determine the direction of transmitting sound waves automatically based on the position of the selected point on the graph that is entering the angle data and the distance of the audience. The transmitted sound is a sinusoidal signal formed from filtered pulse-width modulation (PWM) generated by Arduino Due Microcontroller. The delay time of each signal will be sent to the microcontroller via serial communication. Implementation of graphical user interface design is shown in Fig. 4.

D. Audio beam steering integration

In this section is to integrate or combine system series system and loudspeaker. Both systems are combined to allow users to access the Audio Beam Steering system, as shown in Fig. 5. Overall, the user will determine the horizontal direction on the computer screen to be solved with the sound signal sent by the loudspeaker series based on the object selected by the user by clicking on the object in the GUI app.

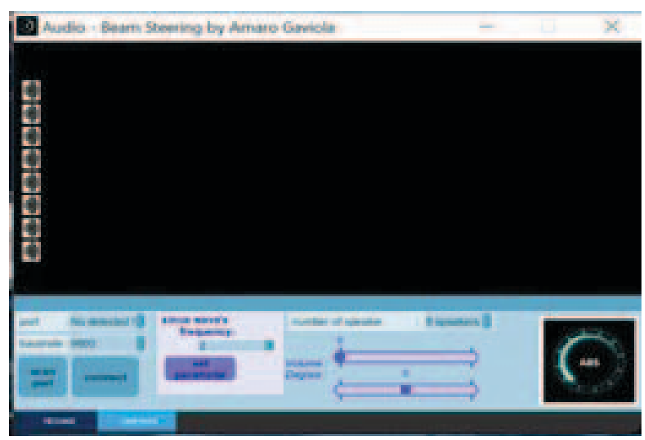


Fig. 4. GUI Display in Audio Beam Steering System

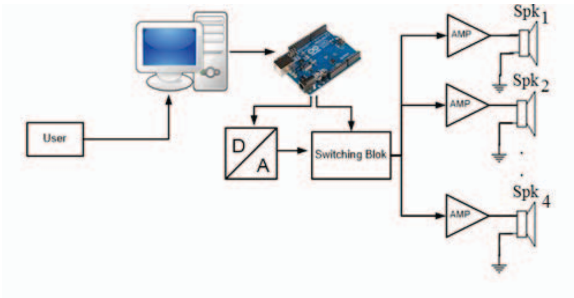


Fig. 5 Diagram of system integration

III. Result

A. *Test audio using microcontroler*

Testing process is conducted to know whether the output signal from microcontroller as expected or not. The Arduino Due microcontroller can generate two PWM signals that can be varied by using Timer1, i.e. on the OC1A and OC1B pins, as shown in Fig. 6. Delays are given on each output pin so that the two signals becomes different in phase, as shown in Fig. 7. Then the filter will be applied to the pins to produce a sinusoidal signal.

The output signal of the microcontroller cannot be directly connected to the speakers, due to the output current limitation. Therefore, the amplifier circuit is used to amplify the output signal from the microcontroller. The amplifier circuit consists of TDA2003 to provide power up to 10 watts RMS. The installation of the amplifiers is shown in Fig. 8.

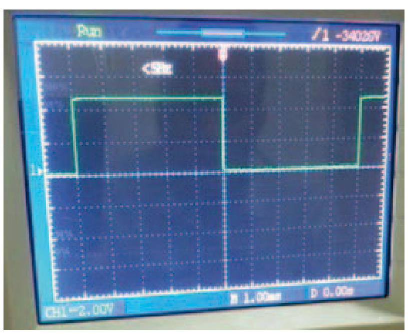


Fig. 6. The PWM signal from the microcontroller

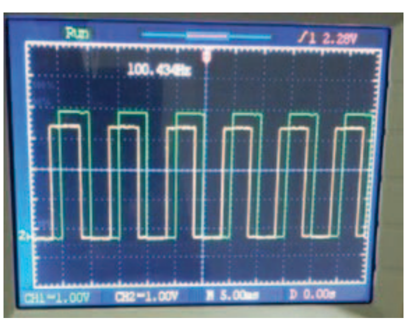


Fig. 7. The delayed signals produced by microcontroller

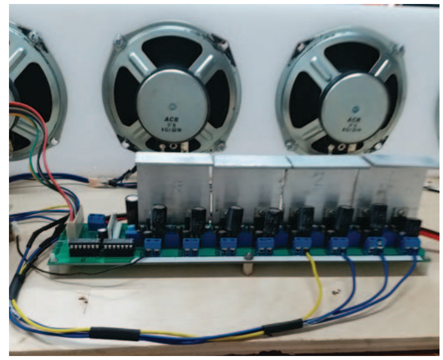


Fig. 8. Installation of the amplifiers

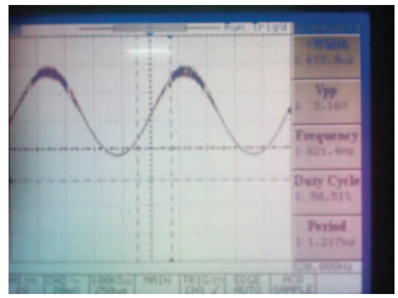


Fig. 9. The sinusoidal signal at the amplifier output

The sinusoidal signal after passing through the amplifier circuit is shown in Figure 9.

B. *Audio Beam Sterring Testing for Audience*

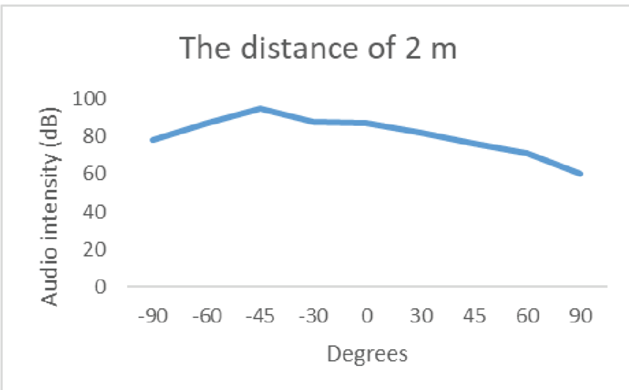
The outdoor test is performed by marking angles as shown in Figure 10. The measurements are conducted in several directions to obtain the lobe patterns describing the level of directivity of the speaker array.

Measuring the lobe patterns that are too close to the speakers will not produce different intensities in any direction. Therefore in this study, the measurements were conducted at the distances of 2 and 5 meters from the speakers. The measurement results for the direction of -45° are shown in Table 1 and Figure 11. It can be seen that the sound intensity for both distances in the direction of the target angle is greater than in the other directions. This shows that the phased array method can steer the audio beam at the desired angle.

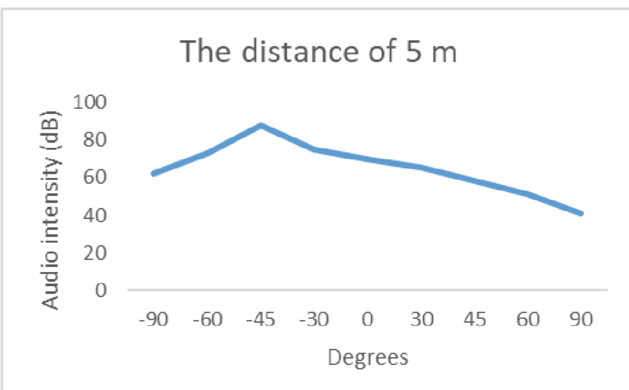

Fig. 10. The measurement of the audio beam steering

TABLE I. THE MEASUREMENT OF AUDIO INTENSITY FOR THE TARGET DIRECTION OF -45° AT THE DISTANCES OF 2, AND 5 METERS

Distance (m)	Angle (°)	Sound Intensity (dB)	Distance (m)	Angle (°)	Sound Intensity (dB)
	-90	78		-90	62
	-60	87		-60	73
	-45	95		-45	88
	-30	88		-30	75
2	0	87	5	0	70
	30	82		30	65
	45	76		45	58
	60	71		60	51
	90	60		90	41



(a)



(b)

Fig. 11 The measurement of audio intensity for the target direction of -45° at the distances of (a) 2, and (b) 5 meters

IV. CONCLUSIONS

In this study, an audio beam steering system from four loudspeaker series has been accomplished with phase difference or phased array method. The computer is used as a Graphical User Interface to determine the direction and distance of the audience who will receive the sound, by generating the delay configuration pattern of each path using Arduino Due Microcontroller. The measurement results for the direction of -45° at the distances of 2 and 5 meters from the speakers showed that the sound intensity generated by the system is greater at target angle than the others. This indicates that the phased array method can steer the audio beam at the desired angle.

ACKNOWLEDGMENT

This research was carried out with financial aid support from the Ministry of Education of the Republic of Timor Leste.

REFERENCES

[1] Anderson B.E., 2006. Grating lobe reduction in transducer arrays through structural filtering of supercritical plates. Ph.D. thesis, The Pennsylvania State University, University Park, PA, pp. 9-26.

[2] Atkins, J., 2010. Optimal spatial sampling for spherical loudspeaker arrays. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 97-100.

[3] Blackstock D.T., 2000. Fundamentals of Physical Acoustics. Wiley, New York, pp. 495-506.

[4] Anderson B. E., Moser G. L., dan Gee K.L., 2012. Loudspeaker Line Array Educational Demonstration. The Journal of the Acoustical society of America, 131 (3), Pt 2, pp. 2394-2400.

[5] Coleman, P., 2014. Loudspeaker Array Processing for Personal Sound Zone Reproduction. Guildford: University of Surrey.

[6] Ishikawa T., Yokoyama S., Harashima N., Takahashi D., Shiozawa J., Yoshino M., Yasuda A., 2014. A Highly Directional Speaker with Amplitude-Phase Control Using a Digitally Direct-Driven System. International Conference on Consumer Electronics, pp. 135-136.

[7] Kinsler L.E., Frey A.R., Coppens A. B., dan Sanders, J.V., 2000. Fundamentals of Acoustics, 4th ed. Wiley, New York, pp. 195-199.

[8] F. Budiman, M. A. Nursyeha, M. Rivai, Suwito, 2016. Bird Voice Recognition Using Mel Frequency Cepstrum Coefficient And Artificial Neural Network On The System Of Bird Human Exposure. National Journal of Electrical Engineering, pp. 64-72.

[9] Szoka, E., 2009. Phased Array Speaker System. Proceedings of the 127th Audio Engineering Society Convention.

[10] Yamada M., Itsuki N., Kinouchi Y., 2004. Adaptive Directivity Control of Speaker Array. International Conference on Control, Automation, Robotics and Vision, pp. 1443-1448.